Portfolio Milestone Module 5: Classification for CubeSat Telemetry Anomaly Detection

Alexander Ricciardi

Colorado State University Global

CSC510: Foundations of Artificial Intelligence

Professor: Dr. Isaac Gang

August 23, 2026

Portfolio Milestone Module 5: Classification for CubeSat Telemetry Anomaly Detection

The proposed CubeSat telemetry anomaly-detection program, from "Portfolio Milestone Module 2: Use-Case Scenario Proposal" (Ricciardi, 2026), will use a supervised binary classification approach to predict whether a telemetry segment indicates normal satellite operation or an anomaly. The classification will provide the first decision point in the CubeSat telemetry program by determining whether a selected telemetry segment represents normal or anomalous behavior. Supervised binary classification is appropriate because the OPSSAT-AD dataset already contains operator-reviewed labels for both classes. Before training, the program will standardize the 18 numerical telemetry features so that measurements with larger numeric ranges do not dominate smaller ones. A fully connected multilayer perceptron, or MLP, with hidden layers of 32, 16, and 8 neurons will then learn patterns that separate normal operating behavior from unusual sensor activity. Rectified Linear Unit activation functions will support nonlinear learning in the hidden layers, while a sigmoid function in the output layer will produce an anomaly probability between 0 and 1. A threshold selected with validation data will convert that probability into a normal or anomalous label. The threshold should not be treated as a claim of certainty. When the probability falls close to the threshold, the program will mark the result as uncertain and gather more evidence before recommending action. The user interface will display the predicted class, anomaly probability, similar labeled telemetry cases, triggered diagnostic rules, and the next recommended checks. This makes the classification result easier to examine instead of presenting a label with no explanation. OPSSAT-AD contains 2,123 labeled segments from nine telemetry channels, and anomalous segments make up only 20% of the dataset (Ruszczak et al., 2025). Because the classes are uneven, accuracy alone could give a misleading impression of performance. The program will also report precision, recall, the F1 score, and a confusion matrix. Recall will show how many real anomalies the model detects, while precision will show how often an anomaly warning is correct. False negatives are especially important because they allow abnormal telemetry to pass as normal, but excessive false positives can also waste an operator's time and reduce trust in the system. “The OPS-SAT benchmark for detecting anomalies in satellite telemetry” (Ruszczak et al., 2025) reported 0.963 precision and 0.929 recall for a fully connected neural-network detector on OPSSAT-AD, which supports the use of an MLP as the initial classifier. Classification will also control the later parts of the hybrid program. A normal result may return the segment to routine monitoring. An anomalous result will activate expert-system rules and start an A* search for a low-cost sequence of diagnostic checks. An uncertain result will cause the similarity-search component to retrieve nearby normal and anomalous cases and will ask the planner to collect additional evidence. Sharda et al. (2024) explain that classification can contribute to knowledge-based models and that predictive results can be combined with prescriptive methods to support decisions. In this program, classification turns raw telemetry into an understandable result, connects that result to evidence and diagnostic actions, and leaves the final judgment with the human operator.

References

Ricciardi, A. (2026). *Portfolio milestone module 2: Use-case scenario proposal* [Unpublished course assignment]. CSC-510 Fundamentals of Artificial Intelligence. Colorado State University Global.

Ruszczak, B., Kotowski, K., Evans, D., & Nalepa, J. (2025). The OPS-SAT benchmark for detecting anomalies in satellite telemetry. *Scientific Data, 12*, Article 710. https://doi.org/10.1038/s41597-025-05035-3

Sharda, R., Delen, D., & Turban, E. (2024). *Business intelligence, analytics, data science, & AI* (5th ed.). Pearson Education.